

WEEK NINE - SHADOW AND SILHOUETTE

Thursday, 3 April 2025, 12:45 - 2:15 PM EST.

Harvard Art Museums 0600

SILHOUETTES PAINTED IN THE SHADOWS were beginnings of Western Art. The earliest European Art are Paleolithic paintings in the Caves of Altamira in Northern Spain. Made 36,000 years ago, but discovered in 1868, the paintings are charcoal and polychrome profiles of bison and other animals. Profiles – silhouettes painted in the shadows – are just enough to convey meaning through representation. Their original purpose is unknown, perhaps they were celebrations or prayers about hunting.



Figure 192: Great Hall of Polychromes of Altamira, published in 1880



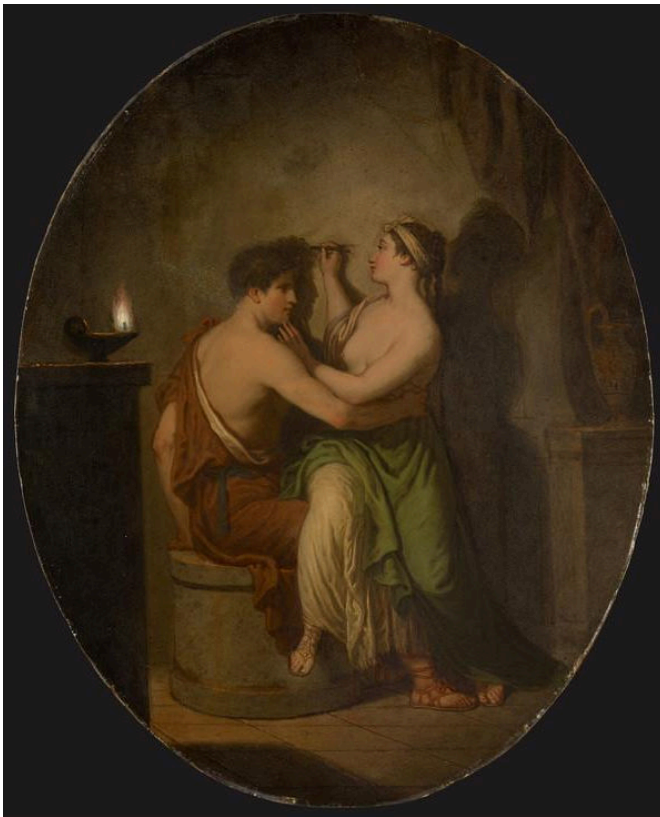
Figure 193: Bison Polychrome, Altamira

Updated: April 13, 2025

GREEK HISTORY gave shadow a role in the beginnings of representational art. Pliny the Elder (AD 23/24 - 79) recorded a story about the origin of portrait sculpture in his *Natural History*. Butades, a potter in Ancient Corinth, noticed a shadow made by lamplight, which his daughter had traced of her boyfriend. Butades filled the outline with clay, inventing the first portrait sculpture.

SHADOW TRACING is an easy-to-use optical effect to help make a realistic representation. This story of Butades was a common theme in European painting, an origin story about Art that reveals a continuing fascination for Greek and Roman antiquity in the 18th and 19th centuries.

PROFILE PORTRAITS of famous Greeks and Romans were most widely seen and most widely distributed on coins. The side profiles is not far from the shadow silhouettes. Side profiles – gaining currency from antiquity and nobility – became an ennobling form of representation.



Updated: April 13, 2025



Figure 194: Alexander the Great, 336-323 BCE. [Harvard Art Museums](#)



Figure 195: Caesar Augustus, Silver coin, 19-18BC. [British Museum](#)



Figure 196: Head of Brutus, Roman Republic, 43 -42 BC. Commemorative coin for Julius Caesar's assassination. [British Museum](#)

Figure 197: David Allan, *The Origin of Painting*, 1775. [Link to painting.](#)

ITALIAN RENAISSANCE artists used shadow, like they used perspective, to add depth. Both modeling shadows and cast shadows helped with the illusion of three-dimensionality. Leonardo da Vinci thought a lot about shadows, devoting 67 pages of his *Codex Urbinas* to their study. But in his notes on painting, *Trattato della Pittura*, he advocated leaving them out.

Light too conspicuously cut off by shadows is exceedingly disapproved of by painters. Hence, to avoid such awkwardness when you depict bodies in open country, do not make your figures appear illuminated by the sun, but contrive a certain amount of mist or of transparent cloud to be placed between the object and the sun and thus, – since the object is not harshly illuminated by the sun – the outlines of the shadows will not clash with the outlines of the lights.

A *SALVATOR MUNDI*, painted in 1500 and attributed to Leonardo, uses shadows for modeling. Its light is diffuse and ambiguous. If this painting were by Leonardo, he followed his own advice. Its attribution to Leonardo made it the most expensive painting ever sold at auction, \$450.3 million to Prince Badr bin Abdulla Al Saud in 2017. It has not been seen since.

ANDREA PREVITALI painted a *Salvator Mundi* in 1519. A commitment to realism is in the strong shadow cast by his raised right hand. Another realistic detail is specular reflection from the glass orb. Leonardo placed a similar orb in Christ's left hand. But both Previtali and Leonardo decline optical realism in the orb itself. Instead of painting an optically-distorting but optically-accurate glass lens, both paint transparent windowpanes.



Figure 198: *Salvator Mundi*, 1500, attributed to Leonardo da Vinci and workshop.



Figure 199: *Salvator Mundi*, 1519, Andrea Previtali. [National Gallery](#)

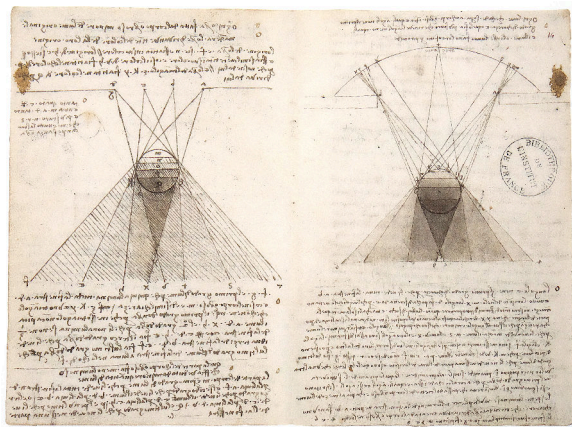


Figure 200: *Study of the Graduations of Shadows on Spheres* from Leonardo's *Codex Ashburnham*

CLASSICAL CHINESE AND JAPANESE ART made scarce use of shadow. *Six Persimmons* by the 13th century monk Mu Chi'I is a centerpiece of Zen art. The artist uses shade without shadows. No shadows for modeling. No cast shadows. Each persimmon is a simple silhouette, each singular in shade.



Mu Ch'I's Persimmons

By Gary Snyder

*There is no remedy for satisfying hunger
other than a painted rice cake.*

—Dōgen, November, 1242.

On a back wall down the hall

lit by a side glass door

is the scroll of Mu Ch'i's great
sumi painting, "Persimmons"

The wind-weights hanging from the
axles hold it still.

The best in the world, I say,
of persimmons.

Perfect statement of emptiness
no other than form
the twig and the stalk still on,
the way they sell them in the
market even now.

The original's in Kyoto at a
lovely Rinzai temple where they
show it once a year

this one's a perfect copy from Benrido
I chose the mounting elements myself
with the advice of the mounter

I hang it every fall.

And now, to these overripe
persimmons
from Mike and Barbara's orchard.
Napkin in hand,
I bend over the sink
suck the sweet orange goop
that's how I like it
gripping a little twig

those painted persimmons

sure cure hunger

Figure 201: *Six Persimmons* by Mu Chi'I, 13th century. [Link to Wikipedia](#)

THE MANNERIST style followed upon the early Italian Renaissance. Mannerism was a direct reaction to da Vinci, Rafael, Michaelangelo and others who strove for symmetry, proportion, and idealized beauty. Mannerists wanted to capture realism, but enhanced for dramatic effect.

GIOVANNI BATTISTA MORONI (~1520 - 1578) used his *Portrait of a Gentleman* to paint an essay on the realistic use of shadow. Here, he studies and compares shadows – a wide variety of cast shadows and modeling shadows – on convex and concave surfaces (Fig. 202).



Figure 202: *Portrait of a Gentleman*, Giovanni Battista Moroni, 1556. [National Gallery](#)

MICHELANGELO CARAVAGGIO (1571-1610) was the most influential Mannerist in using shadow. His followers were called the *tenebrists*, meaning shadowists. Where earlier Renaissance painters sought idealized and sanitized visual truth, Caravaggio insisted on absolute and brutal honesty. His painting of Thomas, the apostle who voiced skepticism about Jesus's resurrection, is a fully realistic depiction of one Bible verse:

Then saith he to Thomas, Reach hither thy finger, and behold my hands; and reach hither thy hand, and thrust it into my side: and be not faithless, but believing.

– John 20:27 (KJV)

CARAVAGGIO creates harsh shadows with dramatic and klieg-like intensity. The scene strives for earthly and tangible naturalism. The holy figures – Jesus, Thomas, and the other apostles – are painted as ordinary humans, worried men with wrinkled brows. Thomas pokes a real finger into a real wound. Caravaggio revolutionized religious depiction by populating biblical scenes with real people.



Figure 203: *The Unbelieving Thomas*, Caravaggio, 1601. [Sarassouci Museum](#)

ITALIAN REALISM made its way to Northern Europe. Rembrandt admired Caravaggio and the tenebrists. A painting from Rembrandt's school is more about shadow than light. Raking light streams through a window and creates an anamorphic shadow stretching across the wall.



Figure 204: *Man Seated Reading at a Table in a Lofty Room*. by Follower of Rembrandt or his school, 1628/30. [Link to painting at the National Gallery.](#)

PIETER JANSSENS ELINGA (1623-1682) was a contemporary of Vermeer. Elinga was as fascinated by shadow as Vermeer was fascinated by light. In this quiet domestic scene, a lady sits at a table, a maid sweeps the floor, and a painter stands in a back room. Shadows – from both direct and reflected light – achieve a complex and virtuosic display of realism.



Figure 205: *Interior with Painter, Woman Reading and Maid Sweeping*, by Pieter Janssens Elinga, 1665-1670. [Link to painting at the Städel Museum, Frankfurt](#)

THE SILHOUETTE was at the beginning of Western Art. But in the United States in the 18th and 19th centuries, it suddenly evolved into a ubiquitous and unique art form. Silhouettes can always be made with scissors and skilled hands. But American technological know-how created gadgets that manufactured high-quality silhouettes at high-speed. Any amateur could make a convincing silhouette. Any customer could afford the ennobling look of a Roman hero.

CHEMICAL PHOTOGRAPHY would end the brief but surging popularity of the American portrait silhouette. But the silhouette retains a timeless quality, and a perspective on other art forms by comparison. When an image is flattened and turned into black and white, all sense of depth and color are lost. The remaining silhouette keeps a sense of realism as pure shadow.

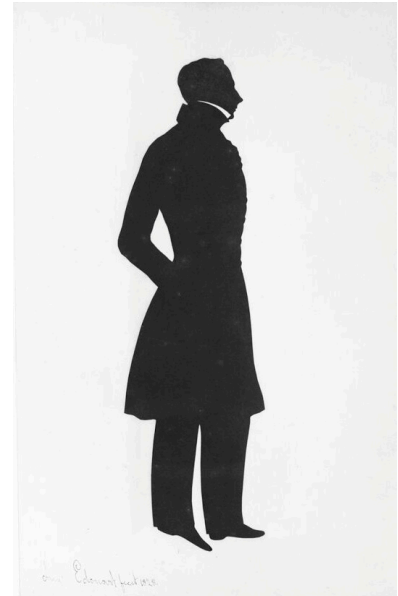


Figure 206: Augustin Amant Constant Fidèle Edouart, French, *Album of Silhouettes*. Harvard Art Museums.

Figure 207: [A Sure and Convenient Machine for Drawing Silhouettes](#) by Thomas Holloway ca. 1788.

THE PHYSIOGNOTRACE by John Hawkins was the major accelerator of silhouette mass-production. With the physiognotrace, anyone could trace a silhouette profile. The silhouette became a democratizing form of art, easy to afford, easy to make. Anyone from American citizen to American President could have a silhouette. The silhouette both equalized and ennobled, art that resonated with the mythic story of American democracy.

SILHOUETTE, a French word that is also connoted with elegance, actually had a less-than-courtly provenance. Étienne de Silhouette (1709-1761) was a French minister famous for being cheap. This elegant but economical form of art was named after a miser.

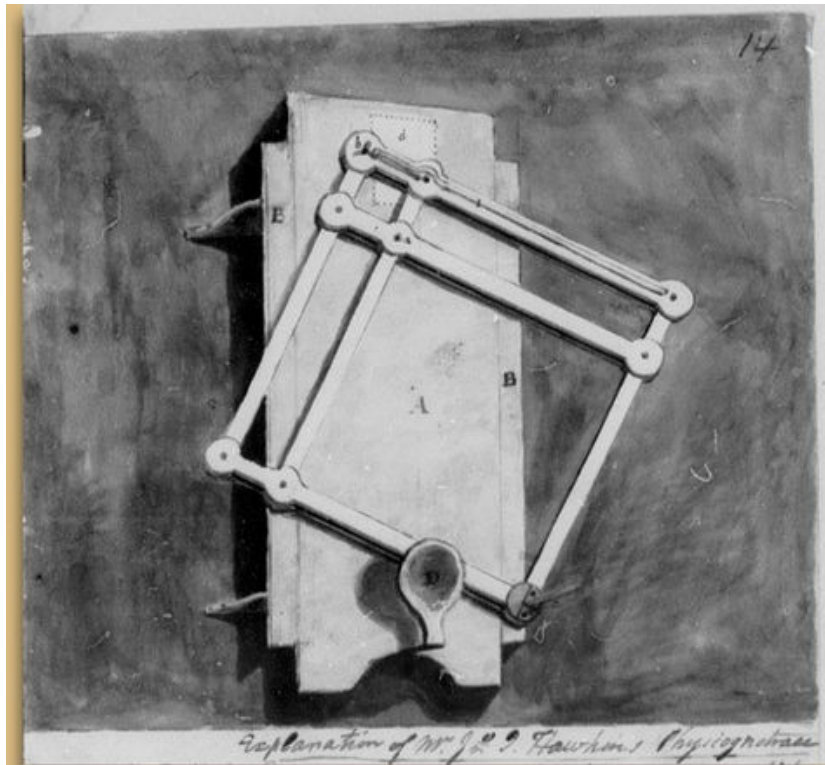


Figure 208: Hollow cut silhouette portrait of George Washington (ca. 1798), by Eleanor Custis, his adopted step-daughter and granddaughter of Martha Washington. [Link to Metropolitan Museum of Art.](#)



Figure 209: Jefferson Portrait by Raphaele Peale 1798. [Monticello Museum.](#)

Figure 210: Drawing of John Hawkins's physiognotrace device.

PHYSIOGNOMIC THEORIES – racism couched in the scientific language of ‘phrenology’ – held that the profile and shape of the skull was an indicator of intelligence and character. A less respectable driver of the popularity of the silhouette was the mistaken idea that it recorded more than a shadow, a rationale of debunked scientific theory.

LA AMISTAD was the slave ship that saw revolt in 1839. It was captured in New York and the case of the slaves was decided in *United States vs. The Amistad* in favor of freedom. However, in John Warner Barber’s *History of the Amistad Captives*, the identities of the 36 captives is shown only in grotesquely reductive silhouette, along with words evoking phrenology.

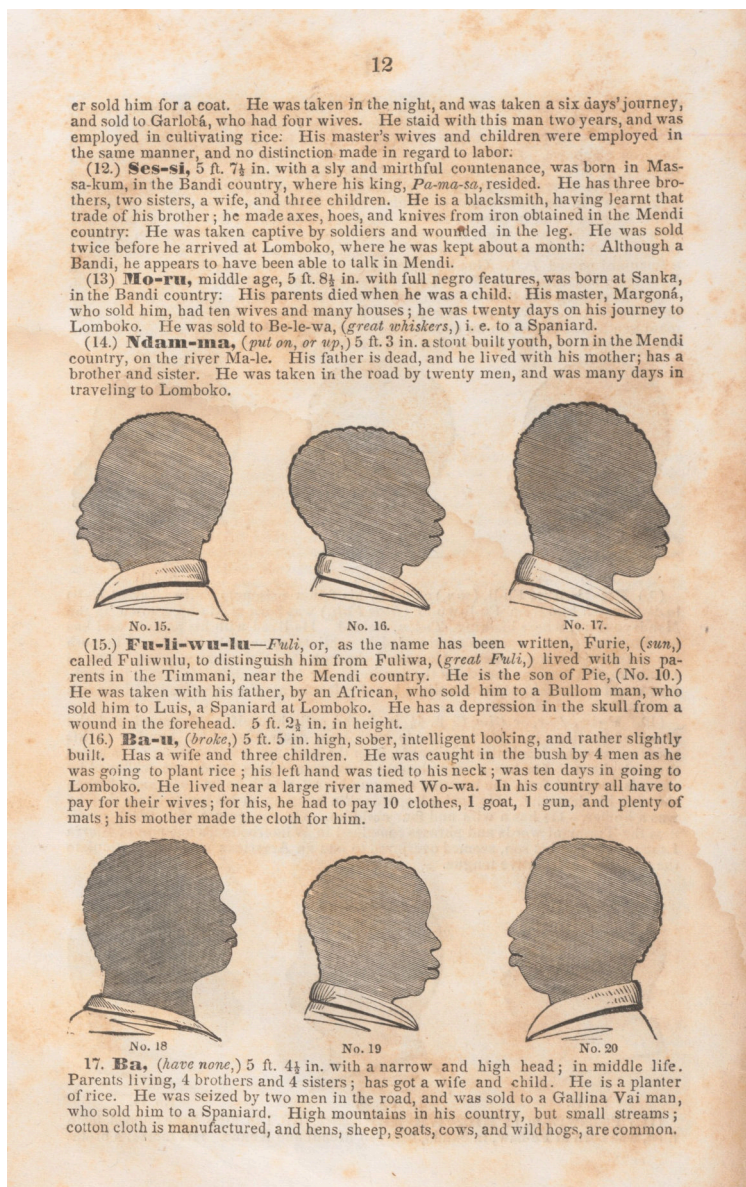


Figure 211: *A History of the Amistad Captives: Being a Circumstantial Account of the Capture of the Spanish Schooner Amistad, by the Africans on Board; Their Voyage, and Capture near Long Island, New York; with Biographical Sketches of Each of the Surviving Africans.* by John Warner Barber, 1840. [Metropolitan Museum](#)

A SILHOUETTE PROFILE can capture identifying detail with precision, but it is also unavoidably anonymizing and ambiguous. When a silhouette is made from any direction but side profile, identity is fully erased. *Skeuomorph* is the name of the iconic blank head that awaits photographs in social media “profiles”.

THE AESTHETIC OF ANONYMITY, using a silhouette not painted in profile, was used by artists. Gerard ter Borch (1617-1681), a contemporary of Vermeer, paints an unknown and unknowable soldier in his grim *Man on Horseback*.



Figure 212: Anonymous



Figure 213: *Man on Horseback* by Gerard ter Borch, 1634. [MFA](#)

CASPAR DAVID FRIEDRICH (1774-1840) painted his *Wanderer Above a Sea of Fog* as a silhouetted figure against a stormy landscape. Here, the anonymity of the silhouette has a more sublime purpose, letting any viewer take the part of the Wanderer. Of painting, Friedrich said:

The painter should paint not only what he has in front of him, but also what he sees inside himself. If he sees nothing within, then he should stop painting what is in front of him.



Figure 214: *Wanderer Above a Sea of Fog* by Caspar David Friedrich, 1817.
[Hamburger Kunsthalle](#)

MODERN ARTISTS have reinterpreted the silhouette as an art form. Kara Walker (1969-) manipulates the history of the silhouette in her disturbing and striking social commentaries (Fig. 216). In contrast to polite portraiture, what the silhouette largely was in the 19th century, Walker evokes exploitation and dark fantasy in 19th century life with her modern use of the art form.



Figure 215: *African/American* by Kara Walker, 1998. [Whitney Museum](#)



Figure 216: *Auntie Walkers Wall Sampler for Savages* by Kara Walker. Kara Walker, a contemporary artist, a mural-scale silhouettes that enacts the cruelty of antebellum plantation life.

KUMI YAMASHITA (1968-) plays with the dimensionality of the silhouette by shaping paper to create silhouettes as shadows when properly illuminated with raking light(Fig. 217). Yamashita's works are anamorphic shadows. The shadows make you think about how they were made.

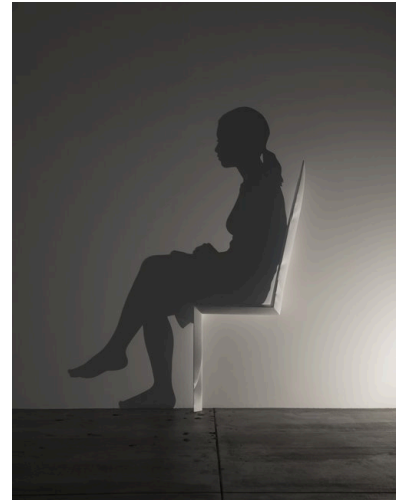


Figure 217: *Chair*, by Kumi Yamashita, 2014. Carved wood, single light source, shadow.

Figure 218: *Fragments*, by Kumi Yamashita, 2009.

FRANCESCA BEWER, the Director of the M-Lab at the Harvard Art Museums, made a surprisingly complex shadow in her photographic still life of lemons.



Figure 219: *Still Life with Oranges* by Francesca Bewer, 2024.

PHOTOGRAPHING

Taking Francesca's photograph as an inspiration, use your phone cameras to capture a surprising or intriguing shadow effect, whether something you stumble upon or something you arrange. Use words to describe what is happening in your photograph.

READING

- "Shades of Black and White" by Penley Knipe. From Asma Naeem. *Black Out: Silhouettes Then and Now*. Princeton University Press, Princeton, New Jersey, 2018. ISBN 069118058X. [Download paper](#)

BIBLIOGRAPHY

- E. H. (Ernst Hans) Gombrich. *Shadows : the depiction of cast shadows in western art*. Yale University Press, New Haven, new edition. edition, 2014. ISBN 9780300210040

Updated: April 13, 2025

WEEK TEN - DRAWING

Thursday, 10 April 2025, 12:45 - 2:15 PM EST.

Visit Ethan Murrow's Studio

59 Wareham Street Boston, MA 02118

DRAWING is the act of pulling an object towards oneself. When the object leaves a mark on a surface, it creates a 'drawing'. The earliest drawings were scratches on surfaces. An engraved stone slab from 11,000 BCE, found in France during the Palaeolithic, depicts part of a horse with a long tail. This plaquette – about 10 cm wide and 3 cm thick – was meant to be portable, shared, and seen.

THE NAZCA LINES in the Peruvian desert are among the largest ancient drawings, over 700 pictures of geometric shapes and animals. The largest are over 300 m, scratched into desert sand between 500 BC and 500 AD. These could not be seen by the artists who made them. Now, they can be seen by airplane or drone camera. They would have required painstaking planning and precision. Presumably they were drawn for the eyes of gods above.

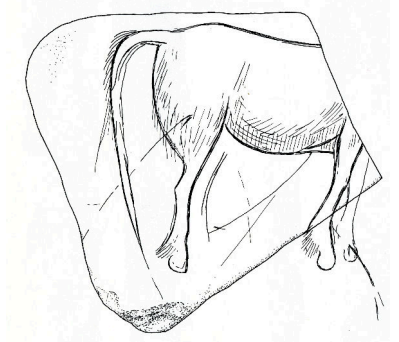


Figure 220: Engraved horse, [British Museum](#)



Figure 221: *The Spider* [Link to Wikipedia](#)

THE INVENTION OF PARCHMENT AND PAPER led to easily portable drawing surfaces and notebooks. At first parchment and paper was rare and expensive. The precious material was reserved for religious worship and text that was “illuminated” by drawings. Pictures were expressions of piety and added visual interest.

THE PRINTING PRESS made paper both widely available and affordable. Soon after Gutenberg’s printing press was reported in 1439, books began to be printed with movable type. Pictures began to be printed with engraved woodblocks and metal plates. Mass production made paper cheap, plentiful, and disposable.



Figure 222: Woodcut showing printer on left removing a page from the press while the printer on right inks a text block. Two printers could produce 3,600 pages a day.

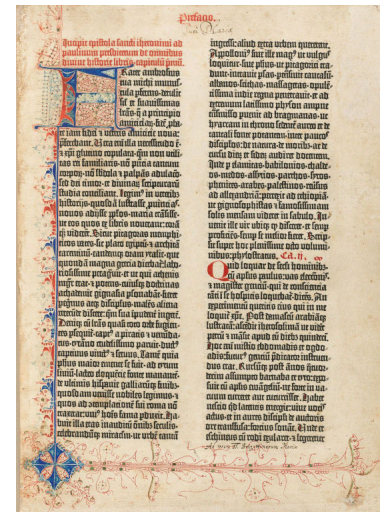


Figure 223: Page from Harvard’s copy of the Gutenberg Bible, printed on paper, one of 23 complete surviving copies. It is always on display in the Harry Elkins Widener Memorial Rooms.

Figure 224: *The Utrecht Psalterbook* was made in the ninth century, a masterpiece with 166 drawings in pen that accompany the psalms and texts. [Wikipedia](#)

WHEN PAPER BECAME PLENTIFUL, it became an efficient way of recording and working out ideas. Paper changed how artists worked, how ideas evolved, and how artistic innovation was communicated. With plentiful paper, Renaissance artists had ample, cheap material to practice and hone their skills. Simply being able to practice was a key turning point in Renaissance accuracy and realism. Before painting or sculpting a work of art, an artist could think, plan, and design with drawing.

PAOLO UCCELLO (1397-1475) worked out ideas about linear perspective on paper. Vasari wrote about Uccello his *Lives of the Most Excellent Painters, Sculptors, and Architects*, a solitary and single-minded man:

gifted with a sophisticated mind, he enjoyed investigating complex mechanisms and the strange products of the art of perspective.

His obsessiveness appears in his calculated drawing of a chalice. But Uccello, like many other artists, considered such drawings as preparatory and disposable. Most are lost.

THE NOTEBOOK was among Leonardo da Vinci's most powerful ideas. We have about 6000 sheets, about a fifth of the estimated total. On these paper sheets, he worked out ideas for paintings like the *Last Supper*. He recorded thoughts and discoveries like the natural flow of water. He tried every drawings tool that was available: pen and ink, metalpoint, charcoal, and chalk.



Updated: April 13, 2025

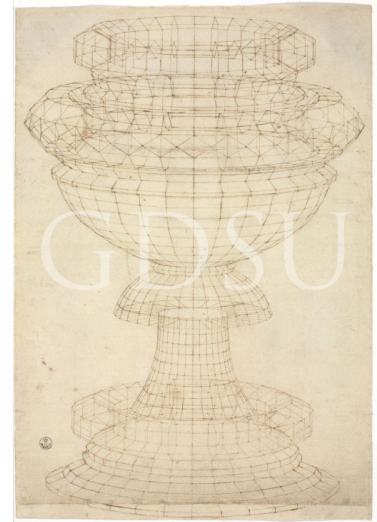


Figure 225: *Studio prospettico di calice*, 1450-70, Paolo Uccello. [Uffizzi](#)



Figure 226: Studies for *The Adoration of the Magi* and *The Last Supper* in brown ink with quill pen and leadpoint, Leonardo da Vinci, [Louvre](#)

Figure 227: *A deluge*, Black chalk, pen and ink, wash. 1517-1518, Leonardo da Vinci. [Royal Collection Trust](#)

MICHAELANGELO BUONAROTTI (1475-1564) started his paintings with drawings, which he usually burned. Vasari writes

to the end that no one might see the labours endured by him and his methods of trying his genius, and that he might not appear less than perfect.

About 600 sheets survive, including studies for the Sistine Chapel. For Adam, he carefully planned his musculature until he found the pose that he wanted. This drawing incorporated Michaelangelo's deep knowledge of the human body, learned from studies of flayed corpses. Michaelangelo's studies led to Adam's convincing pose. This pose was invented from his studies, not drawn from life. It is physically impossible for a human to bend their torso so sharply. These drawings were then enlarged on tracing paper and transferred to the ceiling before Michaelangelo began painting.

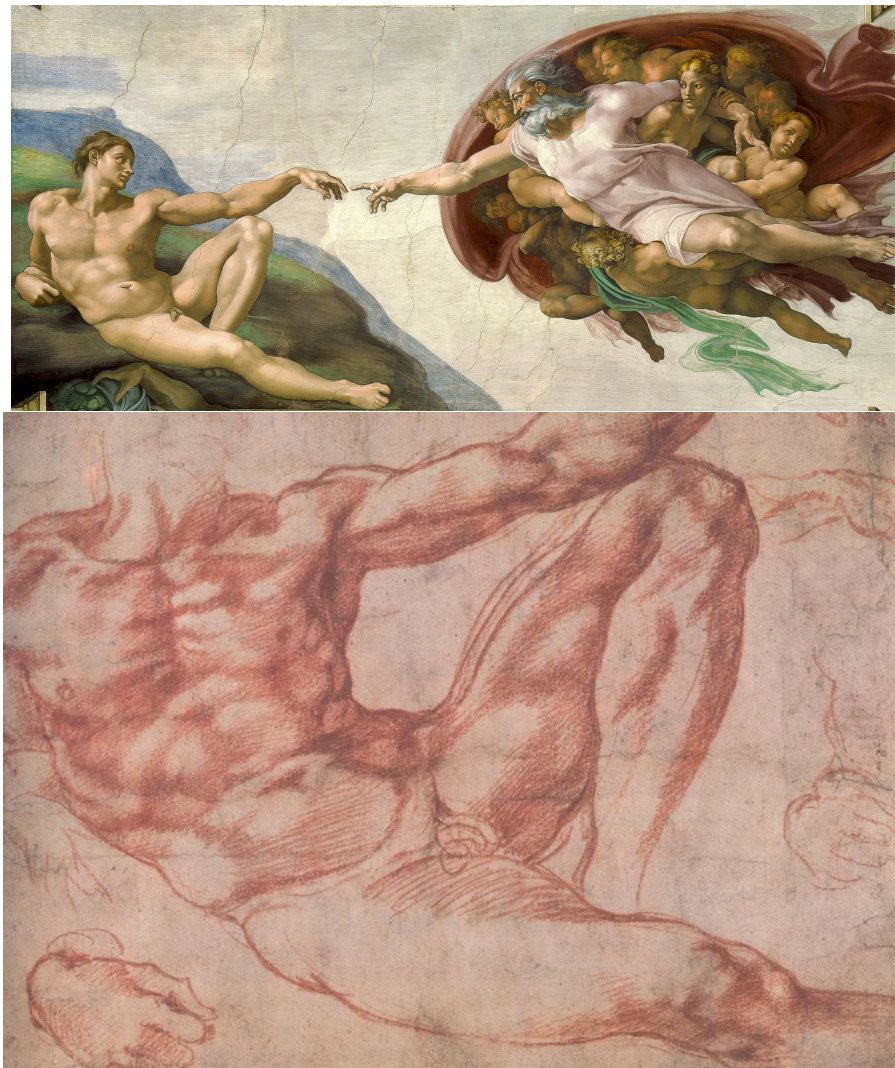


Figure 228: *Creation of Adam* by Michelangelo, 1512 and red chalk study.

SIR PETER PAUL RUBENS (1577-1640) painted grand panoramic scenes that also did not occur. His landscapes were built from ideas recorded on paper as he walked the Flemish countryside. His panel painting of *Het Steen*, the manor where he lived, invents an autumn morning. The painting draws on notes on paper, from drawings and sketches were he recorded effects like the play of light on water. In one drawing, he records an optical phenomenon that he wanted to remember:

The reflection of the trees in the water is browner and more perfect in the water than the trees themselves.



Figure 229: *An Autumn Landscape with a View of Het Steen in the Early Morning* by Rubens, 1636 [National Gallery](#).



Figure 230: *Trees reflected in water at sunset* by Rubens, 1635-38. Pen and brown ink. [British Museum](#)

DRAWING TOOLS IMPROVED OVER TIME. Renaissance tools like ink and metalpoint are unforgiving, impossible to erase. Chalk and charcoal smudged easily. When the graphite pencil was invented in the late 18th century, artists had a new and powerful tool. The point was easily sharpened. Faint and hard lines, texture and shading, could be made with varying pressure and different strokes. The pencil helped turn pure drawings into finished works of art, not just as a step toward other final form like a print or painting.

JOHN CONSTABLE is known for oil landscapes. But his exquisitely detailed graphite drawing of elm trees is a fully finished work of art. To other drawings, he sometimes added watercolor. His attitude to these landscapes, across mediums, is summed up in his quote:

Painting is a science, and should be pursued as an inquiry into the laws of nature. Why, then, may not landscape painting be considered as a branch of natural philosophy, of which pictures are but the experiments?



Figure 231: *A Mill Near Colchester*, 1833, John Constable Watercolor over traces of graphite, with scratching out and a touch of pen and brown ink on the horizon, evidence of squaring for transfer in graphite, with a color wash test verso [Metropolitan Museum](#)

Figure 232: *Elm trees in Old Hall Park, East Bergholt*, 1817, John Constable, graphite on paper. [Victoria and Albert Museum](#)



Figure 233: *A Landscape with a Church Tower and a House* 1809, John Constable, Graphite on cream antique laid paper. [Harvard Art Museums](#)

MANY MODERN ARTISTS now use drawing as their main medium. Ethan Murrow makes large-scale drawings on paper with pencil and murals with paint-pencils. In his work, he looks back on historical narratives, and the ways in which stories are told. Murrow is an admirer of the exquisite naturalistic detail of Jan Van Eyck. In *Passage*, he draws himself boating by a lamb, a quote from *The Ghent Altarpiece*, also called *The Adoration of the Mystic Lamb*.



Figure 234: Images from the Expedia Mural Project in 2019 and from "Hauling" commissioned by the Currier Museum in 2018, by Ethan Murrow.

Updated: April 13, 2025

Figure 235: *Passage*, 2023, Ethan Murrow, graphite on paper, 48"x48".

GIDEON BOK draws the passage of time in confined spaces. He depicts his own his own studio, over and over in various states at different times. By adding a temporal dimension, he makes the room seem alive (Archives of askArt.)



Updated: April 13, 2025

TOBA KHEDOORI often depicting objects from different perspectives. By isolating objects in an unfamiliar setting, she hovers between naturalistic representation and abstraction. Her work is painstaking and precise, slowly working by hand (from www.lacma.org)



Updated: April 13, 2025

PETER WAITE puts structures that we have built into new perspectives. He makes large scale drawings of recognizable places: stadiums, gardens, bridges, museums, monuments, train stations, schools, prisons, casinos, and temples. These are spaces to which we form specific attachments. Looking at these spaces in drawings shifts our perspective. (from www.peterwaite.com)



Updated: April 13, 2025

WATCHING AND WRITING

Watch the movie *Tim's Vermeer*, freely available on the course website or hosted by Claire Swadling. It was by seeing *Tim's Vermeer* that I first wondered how realistic artworks were made, the first inspiration for inventing this course. We used to watch it at the beginning of the class. Instead, we're watching it at the end. In this week's essay, react to the movie with what you now know.

BIBLIOGRAPHY

- Susan Owens. *The story of drawing : an alternative history of art*. Yale University Press, New Haven, 2024. ISBN 9780300277609



Figure 236: *Waiting* c. 1882, by Edgar Degas, pastel on painter. Degas was an oil painter, but as he got older, he drifted toward drawing with pastels. Painting with oil grew tedious, and pastels gave him the strength of painting with the ease of drawing. [Getty Museum](#)

WEEK ELEVEN - ON PHOTOGRAPHY

Thursday, 17 April 2025, 12:45 - 2:15 PM EST.

Updated: April 13, 2025

Visit Abe Morell's Studio
80 Oak Cliff Road. Newton, MA

WILLIAM HENRY FOX TALBOT (1800-1877) wrote that he invented photography after frustration with the camera lucida. He struggled with the optical device to draw pictures of a vacation in Italy in 1833. But the resulting drawings were such that the *"faithless pencil had only left traces on the paper melancholy to behold."* He concluded that the camera lucida only worked for someone who could already draw.

But Talbot was a scientist. In 1831, he had been elected to the Royal Society for mathematics. He also knew chemistry and understood how light interacted with silver salts. When a solution of silver nitrate (AgNO_3) is mixed with sodium chloride (NaCl) and brushed on paper, it forms a layer of light-sensitive silver chloride (AgCl). Light turns AgCl into metallic silver, darkening the paper with illumination, creating a 'negative' image.

Talbot prepared paper with salt solutions in darkness. He exposed the paper to focused images and 'fixed' the result. Fixing means stabilizing the image so that it does not darken with more light. Finally, Talbot reversed the image – turning negative to positive – by exposing a second sheet of paper using the first. A photograph – 'light-writing' from its Greek roots – of a latticed window is the first positive image from a 'negative'.

LOUIS DAGUERRE (1787-1851) found a different way to make photoreactive silver. He treated silver-plated copper plates with fumes that made it light-sensitive. After exposure, an image was made visible with mercury vapor and 'fixed' with chemicals. The resulting *daguerrotype* is a fragile image – a mere layer of atoms on the mirrored surface. When Daguerre announced his invention, Talbot also made his work public.

THE CAMERA OBSCURA was the original housing for the methods of Talbot and Daguerre. A simple lens and box will project a focused image onto a surface. Modern cameras are still called 'cameras' even though they are no longer the 'dark rooms' of Latin translation.



Figure 237: Daguerrotype of Henry Fox Talbot, 1844, by Antoine Claudet

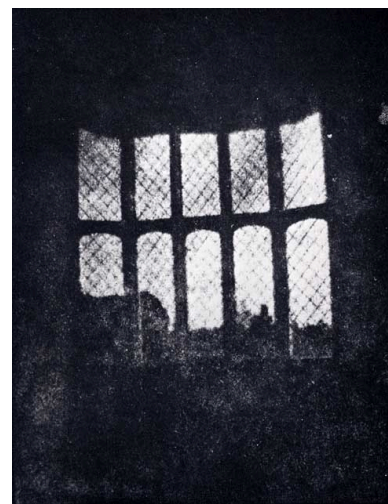


Figure 238: *Latticed window at Lacock Abbey* by Henry Fox Talbot, 1835.



Figure 239: Daguerrotype camera built by La Maison Susse Frères in 1839, with a lens by Charles Chevalier. From Wikipedia

TALBOT was inspired by the camera obscura before he invented the photographic camera. The physics was ancient. Aristotle and da Vinci knew that when rays of light pass through a pinhole, pencils of light illuminate a far surface. Talbot described the first lightbulb of his idea:

This led me to reflect on the inimitable beauty of the pictures of nature's painting which the glass lens of the Camera throws upon the paper in its focus – fairy pictures, creations of a moment, and destined as rapidly to fade away.

It was during these thoughts that the idea occurred to me... how charming it would be if it were possible to cause these natural images to imprint themselves durably and remain fixed upon the paper!

ABE MORELL captures the idea of the photographic process in a photograph.



Figure 240: *Light Bulb* by Abe Morell, 1991, Archival pigment print.

PHOTOGRAPHY WAS INVENTED, or so Talbot thought, to subtract human fallibility from drawing or painting. Early methods of Talbot and Daguerre were quickly replaced by easier-to-use chemistry and devices. The first photographers had Talbot's vision, heightened realism in 19th century landscape and portraiture.

CARLETON WATKINS (1812-1916) was an early American photographer born in Upstate New York. He made 'mammoth plate' albumen prints of Yosemite, positive images the same size as the original negatives. Watkins made the first majestic images of the American West.

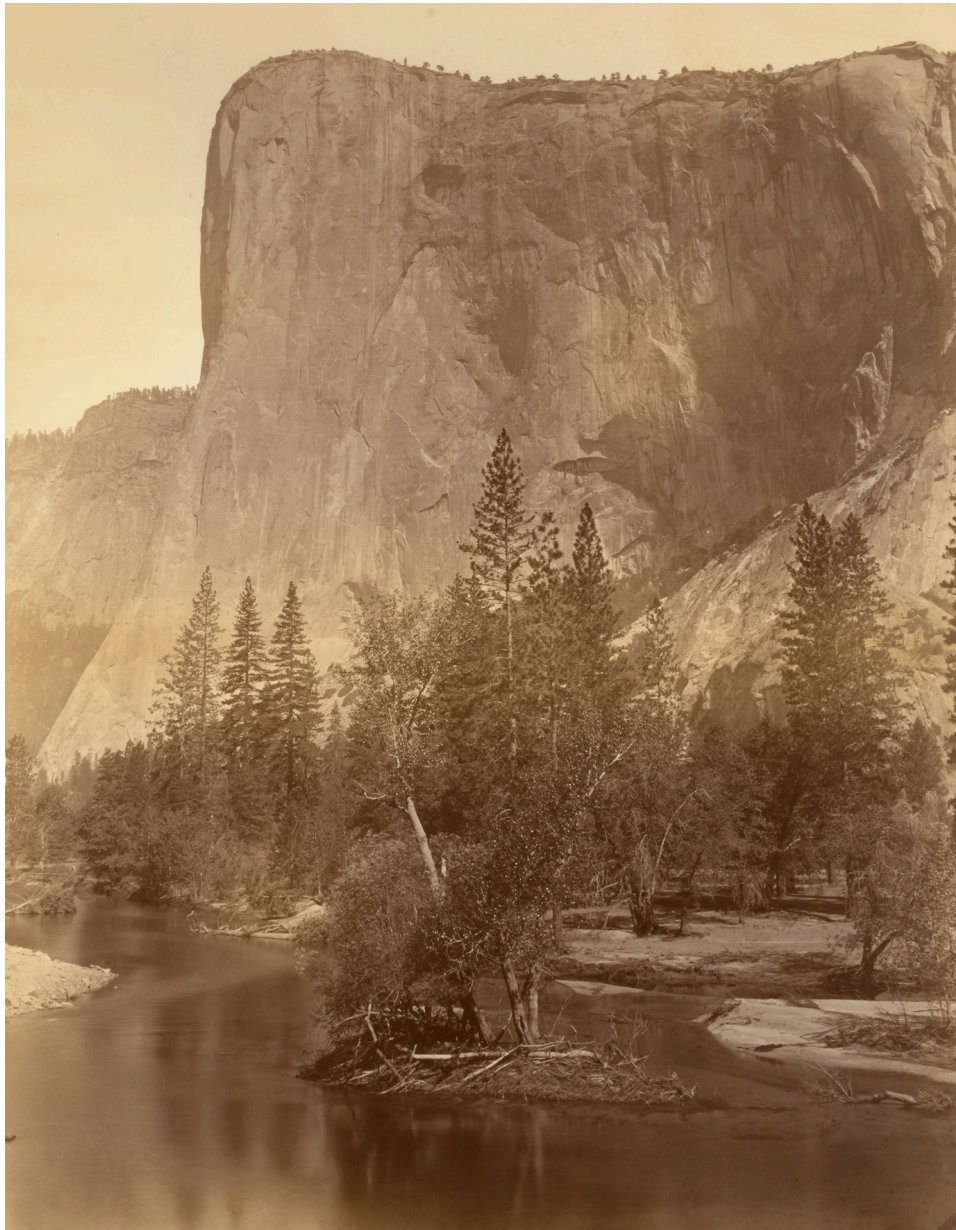


Figure 241: *El Capitan* by Carleton Watkins, 1860s. [Smithsonian American Art Museum](#)

JULIA MARGARET CAMERON (1815-1879) used her camera to make portraits that could have been paintings. She long studied Raphael, Giotto, and Michelangelo from prints that circulated in 19th century England. She took up photography at the age of forty-eight from her home on the Isle of Wight, enlisting models from her family, servants, and neighbors. She wrote:

My aspirations are to ennoble Photography and to secure for it the character and uses of High Art by combining the real and Ideal and sacrificing nothing of the Truth by all possible devotion to Poetry and beauty.

Accordingly, she chose idealized subjects, like photographic illustrations for Tennyson's narrative poetry.



Figure 242: *And Enid Sang*. Albumen Silver Print by Julia Cameron, 1875.

[Getty](#)

And while he waited in the castle court,
The voice of Enid, Yniol's daughter,
rang
Clear through the open casement of the
hall,
Singing; and as the sweet voice of a
bird,
Heard by the lander in a lonely isle,
Moves him to think what kind of bird it
is
That sings so delicately clear, and make
Conjecture of the plumage and the
form;
So the sweet voice of Enid moved
Geraint;
And made him like a man abroad at morn
When first the liquid note beloved of
men
Comes flying over many a windy wave
To Britain, and in April suddenly
Breaks from a coppice gemmed with
green and red,
And he suspends his converse with a
friend,
Or it may be the labour of his hands,
To think or say, 'There is the nightin-
gale;
So fared it with Geraint, who thought
and said,
'Here, by God's grace, is the one voice
for me.'

Alfred Tennyson, *Idylls of the King*

PHOTOGRAPHY HAD NEW POSSIBILITIES with better devices and better chemistry. The first photographic exposures took hours, thus limited pictures to landscape and still life. When exposures took minutes, photography expanded to posed portraiture. With hand-held cameras and sub-second exposure, photography took to the streets.

ALFRED STIEGLITZ (1864-1946) looked for art in urban life. In 1882, he bought his first hand-held camera, a Folmer and Schwing 4x5 plate film camera. The camera let him capture the fleeting moment. In 1893, he took a photograph of steaming horses in front of the Old Post Office in Manhattan.

Naturally there was snow on the ground. A driver in a rubber coat was watering his steaming horses. There seemed to be something related to my deepest feeling in what I saw, and I decided to photograph what was within me.



Figure 243: *The Terminal*. Gelatin Silver Print. Alfred Stieglitz, 1893. [Art Institute of Chicago](#)

PAINTING AND DRAWING changed in subject matter across centuries. Italian artists of the early Renaissance painted idealized images to be worshiped in churches and cathedrals. The Dutch artists of the 17th century painted everyday scenes to decorate homes.

Photography followed a similar arc in changing subject matter. The first generation of photographers pursued idealized visions. Their successors found inspiration in the everyday.

EDWARD STEICHEN photographed a milk bottle on a tenement fire escape in New York in 1915. From his earlier painterly work, he shifted purpose:

When I first became interested in photography, I thought it was the whole cheese. My idea was to have it recognized as one of the fine arts. Today I don't give a hoot in hell about that. The mission of photography is to explain man to man and each man to himself.

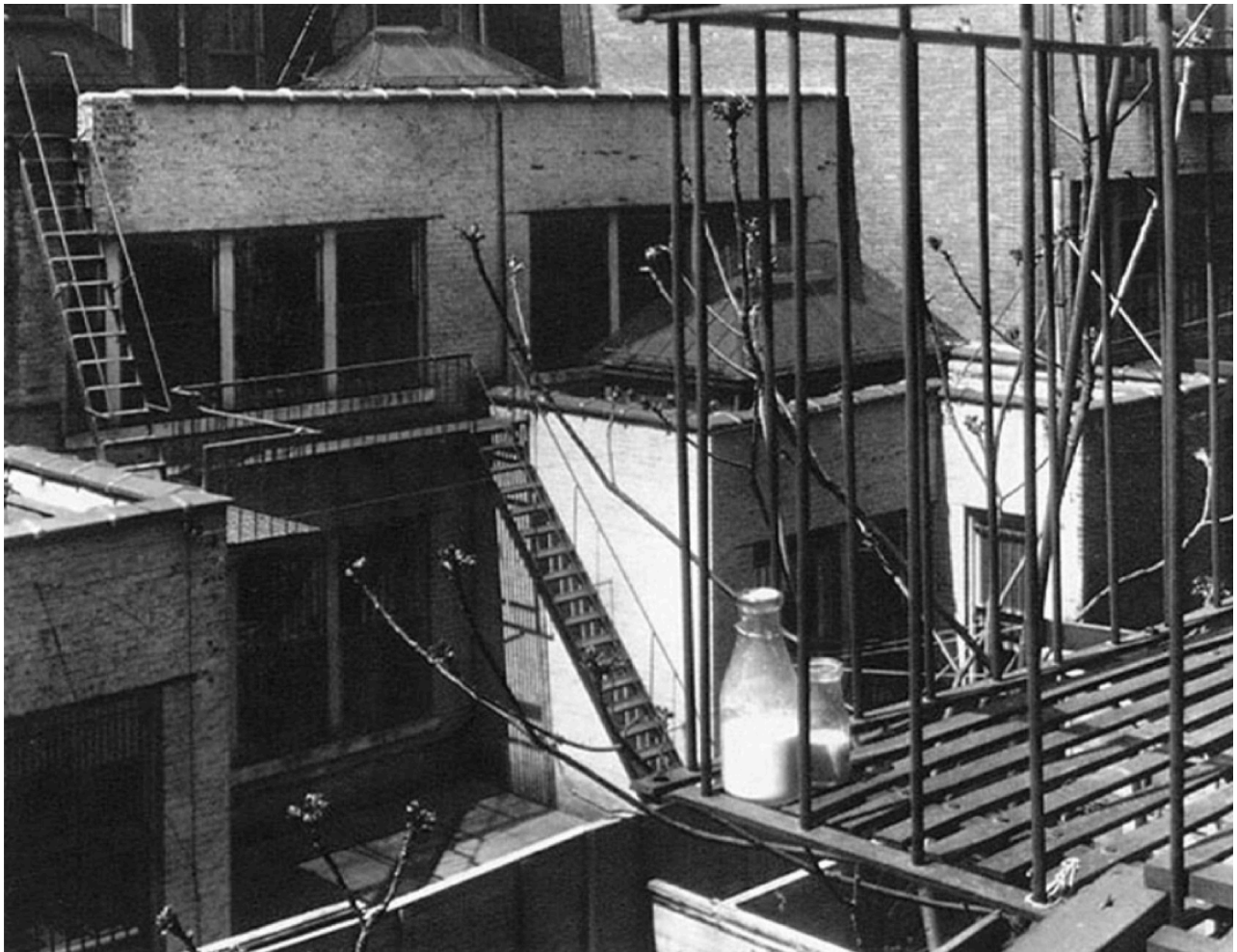


Figure 244: *Milk Bottles on a Fire Escape*.
Edward Steichen, 1915.

DIANE ARBUS (1923-1971) lived and died in New York City. Her photographs capture the human condition without celebration. She connected with people who connected with her, who had not previously been seen by photographers but allowed themselves to be seen by her. Most photographs are full frontal. Her subjects look straight into the camera, looking at Arbus who is looking at them. Because Arbus is looking through a camera, her subjects also look at us looking at them. Arbus's camera turns viewer into voyeur.

Arbus used inexpensive commercial film that allowed one camera to take many photos on a single reel. Arbus could pick among the many photographs of the same scene. Her contact sheet of *Child with a toy hand grenade* shows that she picked the boy's most freakish pose.

Photography does not erase the human from the making of a photograph. Beyond deciding what to photograph, the photographer decides how to enlarge, crop, and print each photograph, which photographs become 'art', and how they will be displayed.



Figure 245: Lady on a busy, NYC, by Diane Arbus, 1957. [The Metropolitan Museum](#)



Figure 246: *Child with a toy hand grenade in Central Park, NYC.*, 1962, Diane Arbus. [The Metropolitan Museum of Art](#)

AS PHOTOGRAPHY CHANGED from the 19th to 20th centuries, painting and drawing also changed, partly in response to photography. When the photograph achieved photographic realism, artisanal art was forced to do what photography could not. The impressionists painted scenes with color instead of shape. The surrealists imagined scenes that had never been seen. The modernists imagined paintings without scenes.

PHOTOGRAPHY SEEMED TO SOLVE THE AGE-OLD PROBLEM of linear perspective with the click of a button. But photography can only project three-dimensional visions onto two-dimensional images. In the 19th and 20th centuries, contemporary scientists were redefining our sense of dimension and space just as artists were redefining art. For example, a square is a two-dimensional shadow of a three-dimensional cube. But the cube is the three-dimensional shadow of a four-dimensional tesseract. Even higher dimension can be imagined, but must be flattened onto two-dimensional images to be seen as pictures. In *Corpus Hypercubus*, the Surrealist Salvador Dalí (1904-1989) depicts Christ crucified on a tesseract.

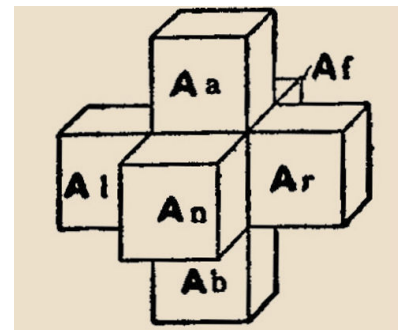


Figure 247: *Hypercube* by Charles Hinton, 1912.

Figure 248: *Crucifixion (Corpus Hypercubus)* by Salvador Dalí, 1954. [Link to Metropolitan Museum of Art.](#)

SUSAN SONTAG (1933-2004) wrote *On Photography*, her seminal meditation on the camera's effect on modern visual culture. We live in a world where everything that can be photographed seems to have been photographed. But the visual reach of photography and art is not infinite. We see what we are shown.

THE ALLEGORY OF THE CAVE in Plato's *Republic* is an older metaphor about the limited reach of visual experience. Socrates and Glaucon have a dialogue about an unfortunate people who are chained and imprisoned inside a cave. The heads of these unfortunates are fixed so that they only see one wall inside the cave. Shadows are projected onto this wall from an outside world that is unseen by the imprisoned. Free people live in the unseen world, carrying puppets and using fire to cast shadows into the cave. For the unfortunate prisoners, the shadows in the cave are their entire and only visual reality. Plato's metaphor describes the human predicament. We see what we are given to see, shadows of larger and hidden realities.

PLATO'S CAVE, in another sense, was the eye-opening device of the first movie theaters. In an etching, Samuel van Hoogstraten (1627-1678) depicts a shadow play in 17th Amsterdam, where silhouettes are projected and enlarged on walls. Hoogstraten's viewers are voluntarily entertained by actors.

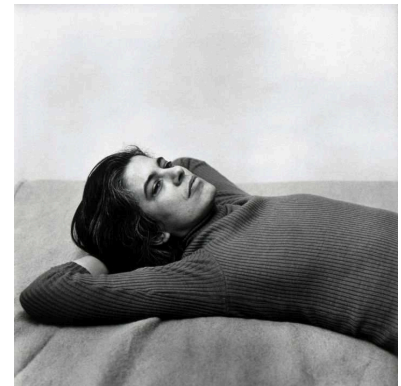


Figure 249: Susan Sontag by Peter Hujar, 1975. [Metropolitan Museum of Art](#)

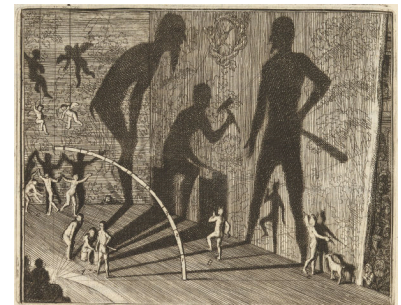


Figure 250: *Inleyding tot de Hooge Schoole der Schilderkonst* by Samuel van Hoogstraten, 1678. [National Gallery of Art](#)

Figure 251: *Plato's Cave* by Jan Saenredaam. 1604. [British Museum](#)

MODERN PHOTOGRAPHERS have found new ways to challenge or use Plato's Cave. The movie theater is our modern cave, where viewers seat themselves to see what the movie wants us to see.

HIROSHI SUGIMOTO (1948-) brought his own camera inside a cinema. When the movie started, he opened the shutter of his large-format camera, exposing the film for the length of the movie. The resulting image stretches the dimension of time.



Figure 252: *Paramount Theatre, Oakland*
by Hiroshi Sugimoto, 1994. [MFA](#)

HIROSHI SUGIMOTO is also known for seascapes that flatten the dimensions of space. Here, the only horizontal is the horizon. After seeing Sugimoto's seascapes, the horizontals in any Rothko begin to look like horizons. Sugimoto achieves painterly abstraction with photographic realism.



Figure 253: *Ionian Sea, Santa Cesarea I*
Sugimoto Hiroshi, 1990 [MFA](#)

ABE MORELL (1948-) turned an ordinary hotel room into Plato's Cave. He blacked out the windows. Light could only enter through a pinhole in the curtain. He opened the shutter of a camera that gradually captured the little light that entered the room and reflected from its surfaces. Because so little light enters the pinhole, long exposures were needed.



Figure 254: *Camera obscura image of the Empire State Building in Bedroom* by Abe Morell, 1994. [MFA](#)

DIGITAL CAMERAS now allow Abe Morell to work much more quickly. An ultrasensitive, high pixel density camera can capture images in minutes what used to take hours with chemical photography. A massive inkjet printer enlarges each digital image.



Figure 255: *View of Lower Manhattan, Sunrise* by Abe Morell, 2022. [National Gallery of Art](#)

A TENT CAMERA OBSCURA took Abe's camera to the landscape. His device uses a periscope to project images of a surrounding landscape onto the ground inside his tent. A digital camera then captures the projected image projected, leading to surprising painterly effects. Abe has taken photographs in Monet's Gardens in Giverny, Van Gogh's fields in Arles, and Constable's English landscapes. With his "camera inside a camera obscura", Abe's photographs come full circle with Talbot's hopes for the photographic camera as painterly device.

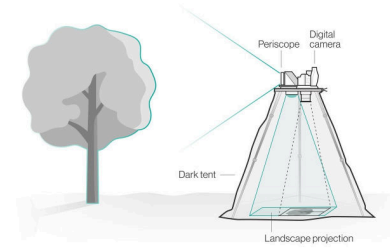


Figure 256: *Water Lilies in Monet's Garden* by Abe Morell, 2023, taken with tent camera obscura.

LOOKING

- Abe Morell's Slide Show. [Link to slides](#)

WRITING – ESSAY TEN

Susan Sontag's essay *Plato's Cave* was written in the 1970s. Her incisive commentary describes an explosion in visual culture caused by the hand-held camera and commercial photography. Today, we are witnessing another, perhaps larger, convulsion in visual culture. We all carry smartphones with powerful cameras. Does Sontag's critique ring true? Are her observations amplified by smartphones and internet culture? Or are they quaint and obsolete?

READING

- Read "Plato's Cave" from *On Photography* by Susan Sontag [Download book](#)

BIBLIOGRAPHY

- Abelardo Morell. *Camera obscura*. Bulfinch Press, New York, NY, 1st ed. edition, 2004. ISBN 0821277510
- Susan Sontag. *Susan Sontag : essays of the 1960s and 70s*. Library of America 246. Library Of America, New York, NY, 2013. ISBN 9781598532555
- Martin Kemp. *Seen/unseen : art, science, and intuition from Leonardo to the Hubble telescope*. Oxford University Press, Oxford ; New York, 2006a. ISBN 0199295727



Figure 257: *Poppy Field #s, Near Vetheuil, France* by Abe Morell, 2022, taken with tent camera obscura.